Root Zone KSK Rollover – FAQ

What is the root zone key signing key?

- The root zone key signing key (KSK) is a cryptographic public-private key pair that plays an important role in the Domain Name System Security Extensions (DNSSEC). The root zone KSK serves as the trusted starting point for DNSSEC validation, like how the root zone serves as the starting point for DNS resolution.
- Just as one starts at the root zone to resolve a domain name anywhere in DNS, software performing DNSSEC validation trusts the root zone KSK and builds a “chain of trust” of successive keys and signatures to validate the authenticity of any signed data in DNS.

What does the root zone key signing key rollover involve?

- This root zone KSK rollover process replaces the existing root zone trust anchor (KSK-2017) with a new root zone trust anchor (KSK-2024).

Why roll over the root zone KSK?

- It is not good for a cryptographic key to remain in use for too long. Like any password, it needs to be changed sometimes.
- It is better to make proactive changes during normal operations when things are running smoothly, rather than react in an emergency.

Who needs to know about the root zone KSK rollover?

- Internet service providers, enterprise network operators, and others who operate DNSSEC validation must update their systems with the public part of the new key signing key.

How will they know?

- ICANN is executing an extensive outreach campaign to ensure that those who currently use the KSK know about the change.
- Interested parties can learn more on the ICANN and IANA websites, where they can also follow KSK updates and join a special mailing list. They can also follow ICANN on social media to stay informed.

What is the impact on Internet users?

If completed smoothly, there will be no perceptible change for end users.

What could go wrong?

- It is possible that some software performing DNSSEC validation will not be updated with the new KSK, or that some systems may not properly handle changes to the trust anchor file published on the IANA website.
- If these issues become widespread, the Root Zone Management Partners could decide to reverse the changes and return the system to a stable state. This is known as a “back-out scenario.”
- The rollover timeline may also be adjusted if additional time is needed to help ensure the stability and reliability of DNSSEC validation during the process.

What would the effect be of a back out or extension?

- The goal of a back out or extension would be to maintain operational stability, so the effect on end users would be minimal.

How long would a back out or extension last?

- It could last indefinitely or until the causes that led to a back out are studied and rectified. The fixes would then be folded into a new root KSK rollover process.

Will network operators see a financial impact from the root zone KSK rollover?

- In most cases, preparing for the root zone KSK rollover will have a small associated cost. However, network operators with dedicated IT staffers can help prepare for the root zone KSK rollover during routine network maintenance, without additional costs.
- After a network operator prepares for the rollover, there is likely to be minimal or no further financial impact. Network operators that verify their DNS resolution infrastructure can support the automated update of the key and will not need to make any changes to their infrastructure or modifications to their operational procedures.
- If network operators are unprepared for the rollover and they have enabled DNSSEC, they run the risk of a significant financial impact. If the trust anchor has not been updated to reflect the new key, DNS resolvers will treat responses signed under the new key as having been tampered with and will discard those responses. This will

result in end users getting an error any time they look up a domain name, which could result in support calls from customers.

How exactly will the root zone KSK be rolled over?

- The root zone KSK rollover is performed gradually through a structured, multiphase life cycle designed to ensure continuity of DNSSEC validation.
- A KSK life cycle consists of eight phases over approximately six years. The key remains actively used for signing for about three years, and each phase is tied to a scheduled key ceremony.
- During these ceremonies, the KSK signs the DNSKEY Resource Record Set (RRset) for the root zone. These signatures become part of Signed Key Responses, which support the DNSSEC chain of trust used by validating resolvers.

What is the timing of the eight phases?

- The rollover process is intentionally gradual and spans several years to help ensure DNSSEC validation remains stable across the Internet.
- KSK-2017 has been the active root zone key signing key since 2017. As part of the rollover process, the successor key, KSK-2024, was generated in 2024 and published to the root zone in February 2025. Since then, both keys have been actively published in the DNS leading up to KSK-2024 taking over signing duties on 11 October 2026, giving validating resolvers time to update their trust anchors and prepare for the transition.
- The actual rollover – when KSK-2024 becomes the only active signing key – is currently planned for the fourth quarter of 2026.
- The following timeline describes the introduction of the successor KSK followed by the decommissioning of the predecessor KSK.

Phase A: Key generation (April 2024)
KSK-2024 is generated at the first key management facility.
Phase B: Key replication (July 2024)
KSK-2024 is copied to the second key management facility. The KSK now is qualified for entering the production state.
Phase C: First data is signed with KSK-2024 for use in **Phase D** (October 2024)
The first set of key signing requests are signed.
Phase D: Publication (February 2025)
KSK-2024 is published to the root zone.
Phase E: Rollover (Q4 2026)
Only KSK-2024 is used to sign the root zone.
Phase F: Revocation (Q1 2027)
KSK-2017 is removed from the root zone.
Phase G: Delete 1 (Q2 2027)

KSK-2017 is deleted from the first key management facility.

Phase H: Delete 2 (Q3 2027)
KSK-2017 is deleted from the second key management facility.

What action can be taken now?

- Software developers that create or maintain DNSSEC validation software should make certain it conforms to RFC 5011.
- For software that does not conform to RFC 5011, or software that is configured to not use RFC 5011, a publication stream trust anchor file will be available on the IANA website. The file should be retrieved when the resolver starts up, and when the KSKs in the DNSKEY RRset in the DNS root zone are changed.
- Software developers and operators of validating resolvers can access operational tests, developed by ICANN, to evaluate whether their systems properly implement RFC 5011 and will automatically update during the root zone KSK rollover.

How can I get involved?

Ask a Question
Send an email to globalsupport@icann.org with "KSK Rollover" in the subject line to submit your questions.

Join the KSK Rollover Discussion List
Sign up to the mailing list for public discussions on related issues.